

21. Show that if n is an odd integer, then there is a unique integer K such that n is the sum of $(K-2)$ and $(K+3)$.

Solution: **Existence:** By Defn. 1, $\exists$ integer K such that $n = 2K+1$
 $\Rightarrow n = 2K + (-2+3) = (K+K) + (-2+3) = ((K+K) + (-2)) + 3$ (Associative Law)
 $= (K + (K+(-2))) + 3$ (Associative Law) $= ((K+(-2)) + K) + 3$ (Commutative Law)
 $= (K+(-2)) + (K+3)$ (Associative Law)

Uniqueness: Let, $\exists$ integers K and K_1 such that $n = (K-2) + (K+3)$
 $= (K_1-2) + (K_1+3)$
 $\Rightarrow n = (2K+1) = (2K_1+1)$. By Defn. 1, it is consistent with the fact that n is an odd integer. $\therefore 2K = 2K_1 \Rightarrow 2(K-K_1) = 0$. $\therefore K-K_1 = 0 \Rightarrow K = K_1$ $\square$

22. Prove that given a real number x there exist unique numbers n and ε such that $x = n + \varepsilon$, n is an integer, and $0 \leq \varepsilon < 1$.

Solution: We know that for any arbitrary real number x , there exists an integer n s.t. $x-1 < n \leq x$. Our goal is to show that n is unique.

Let, $\exists$ an integer n_1 s.t. $x-1 < n_1 \leq x$. $\therefore (x-1) - (x-1) < n - n_1 \leq x - x$
 $\therefore 0 < n - n_1 \leq 0$. This can only happen if $n - n_1 = 0 \Rightarrow n = n_1$.

Let, $x = n + \varepsilon$. $n \leq n + \varepsilon \Rightarrow 0 \leq \varepsilon$. $x-1 < n \Rightarrow x-1 < n+1 \Rightarrow n+\varepsilon < n+1$
 $\Rightarrow \varepsilon < 1$. $\therefore 0 \leq \varepsilon < 1$. We shown the existence of an ε in $[0, 1)$ s.t. $x = n + \varepsilon$.

Now, we shall show the uniqueness of ε .

Let, $x = n + \varepsilon_1$. It's easy to show that $0 \leq \varepsilon_1 < 1$.
 $\therefore n + \varepsilon_1 = n + \varepsilon \Rightarrow \varepsilon_1 = \varepsilon$ $\square$